



• **Q** Epsilon waves +

• **Q** Arrhythmogenic right ventricular dysplasia.

- Osbourne wave - Seen in hypothermia - $< 35^{\circ}\text{C}$ (in downswing of R wave)
- Delta wave - Seen in WPW syndrome (in upswing of R wave)

Q. A short-stature female presents with a history of wearing socks in the summer season. Physical examination shows icy cold toes with cyanosis. CXR done shows?

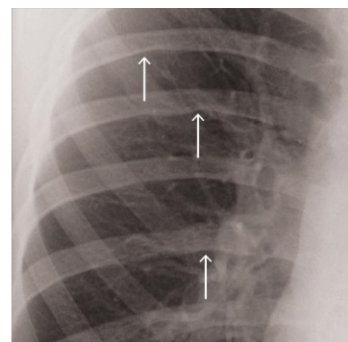
01:06:08

- Hyperparathyroidism
- Neurofibromatosis
- Coarctation of aorta
- Multiple myeloma

Ans ©

INFERENCE -

- Short stature female - possibility of Turner syndrome
- Icy cold toes - obstructive lesion +
- **CXR** - Inferior rib notching (d/t dilated intercostal artery pulsation)
- Commonest cardiac vascular lesion in Turner's:
 - Bicuspid aortic valve > Coarctation of aorta
 - Coarctation of aorta → Erosion of lower border of ribs due to intercostal artery tortuosity



Q. CXR was done for a 15-year-old boy with low cardiac output. Which is incorrect about the patient? 01:08:37

- Reverse split S1
- Straightening of left heart border
- Carey Coombs murmur
- Seagull murmur

Ans (d)

INFERENCE -

- CXR- Straightening of left heart border - Mitral stenosis
 - As obstruction worsens - Narrow split S1/single S1
 - If T/t not given - Reverse split S1
 - Leading cause of mitral stenosis → Rheumatic heart disease → Carey Coombs murmur +
- **Seagull murmur** - seen in mild aortic regurgitation





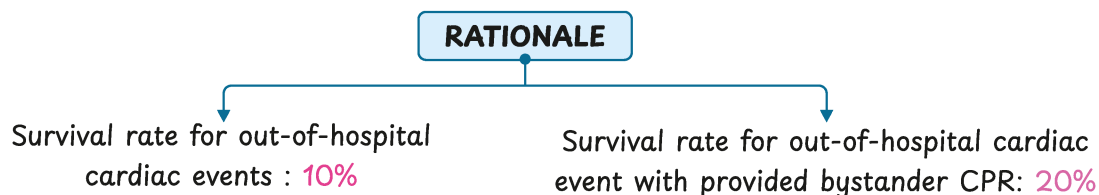
12.3 BASIC LIFE SUPPORT



BASIC LIFE SUPPORT

RATIONALE FOR BLS

00:00:21



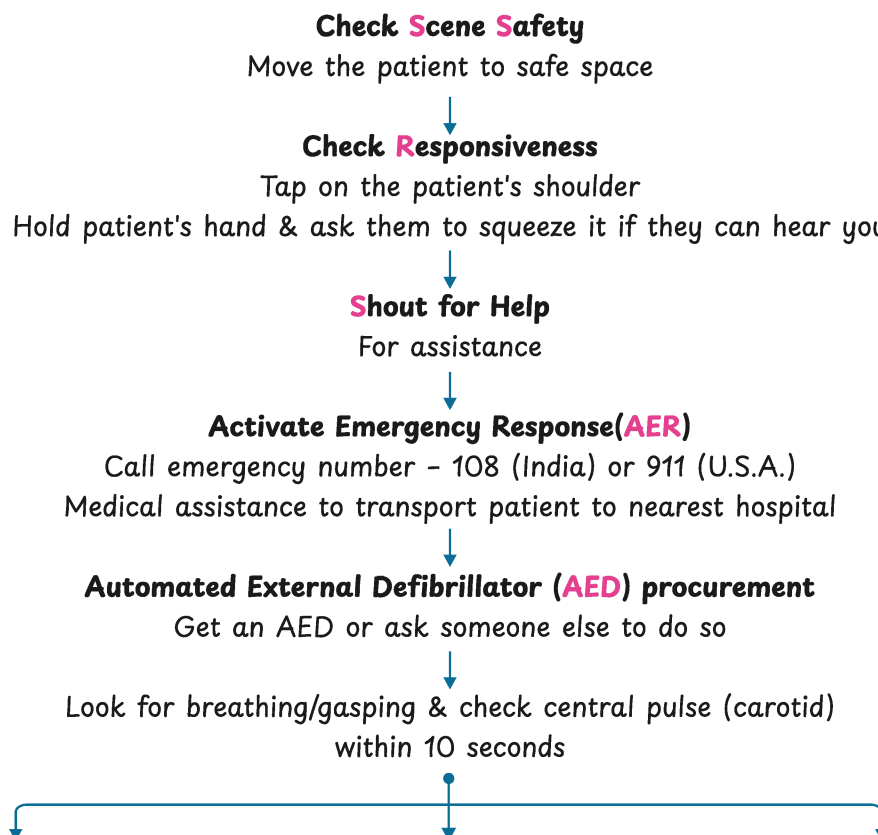
OUT OF HOSPITAL CARDIAC ARREST ASSESSMENT

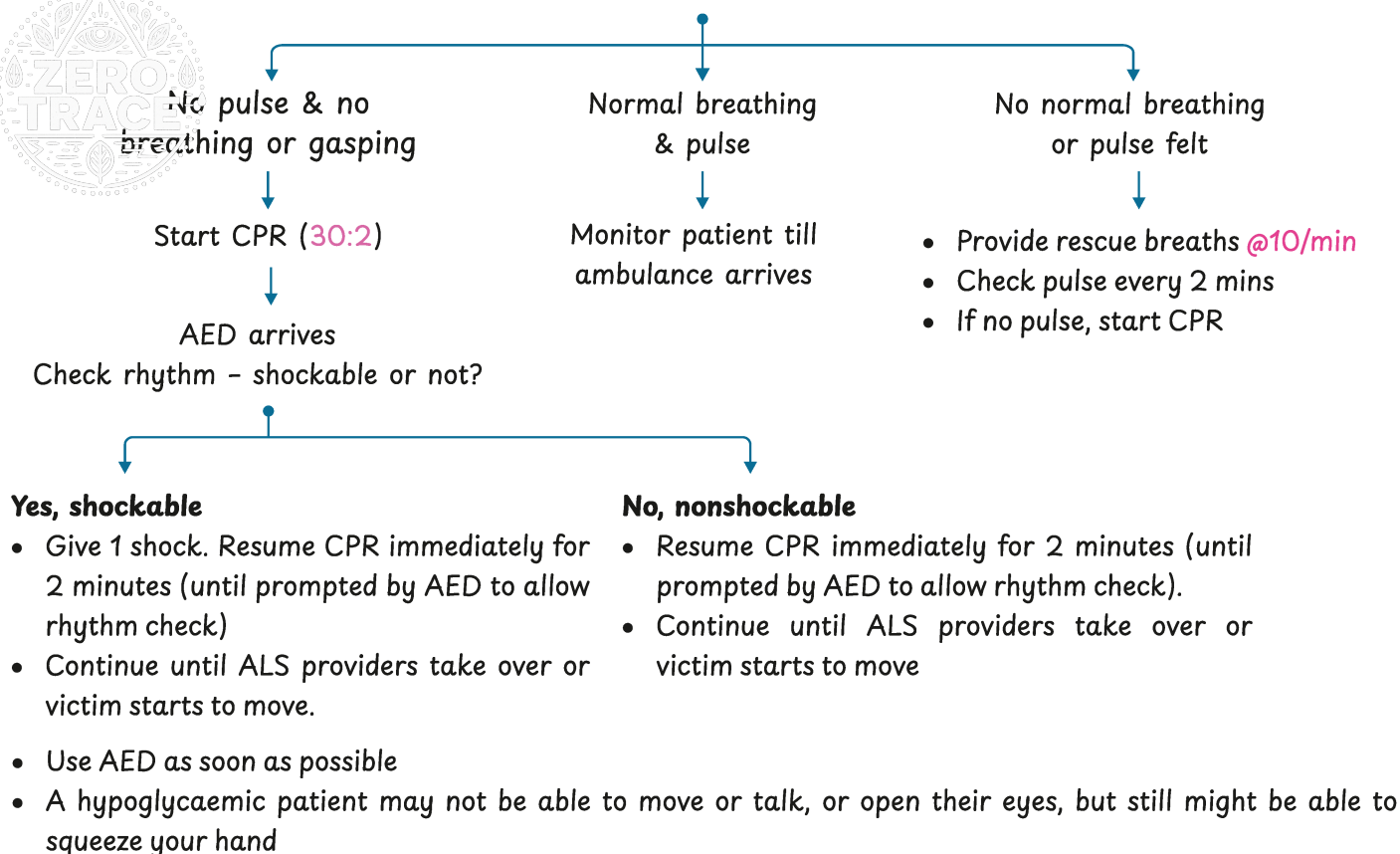
- Unresponsive
- Pulseless (Both peripheral & central)
- Apnea/Gasping/Agonal rhythm
- Leading cause of cardiac arrest: Ischemic heart disease (IHD)/Coronary artery disease(CAD)

CASE SCENARIO

00:03:57

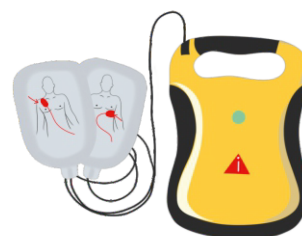
- You are travelling for a medical conference & an elderly male of 60 years suddenly collapses at the airport boarding gate in front of you. What is the sequence of steps to be taken to manage this patient?





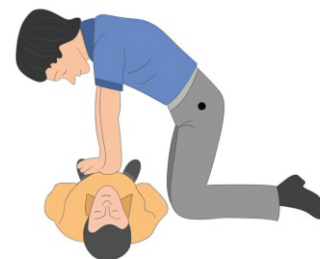
AUTOMATED EXTERNAL DEFIBRILLATOR

- Shockable rhythm - DC shock of 200 J
- Position of pads:
 - Right pad: Right upper sternal border, 2nd intercostal space
 - Left pad: Lateral to apex of heart, 5th intercostal space
→ Middle of pad in mid-axillary line



CARDIOPULMONARY RESUSCITATION (CPR)

- Ratio of Chest compressions to rescue breaths - 30:2
- One cycle of CPR - 2 minutes (Position swap between rescuers)
- Chest compressions rate @ 100-120/min
 - 2 chest compressions in 1 second
- Site of chest compressions: Lower 1/3rd of sternum
 - Do not press xiphisternum as it is pointed & can cause damage to liver
- Depth of chest compressions: 5-6 cm
 - Allow for elastic recoil chest
 - Allow for coronary flow during diastole
- Fulcrum of movement - Hip joint
- M/c rib fractured during CPR - 4-6th ribs (more angulated)



AIRWAY MANAGEMENT

- Bag & mask ventilation - AMBU (Artificial Manual Breathing Unit) bag

00:21:26

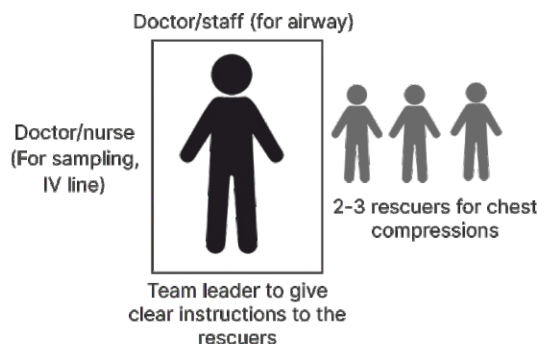
RATIO OF CHEST COMPRESSIONS TO RESCUE BREATH AGE-WISE

AGE GROUP	RATIO	NUMBER OF RESCUERS
Child	30:2	1
Child	15:2	2
Adult	30:2	1-2
Neonate	3:1	2

- Head tilt & chin lift: **Sniffing position**
 - Better access to airway
- Mask held between thumb and index finger, forming 'C' shape, allowing mask to fit tightly to face
- Remaining three fingers placed on lower border of mandible/chin, allowing for chin lift
- Even if it is done in emergency department of government hospital, it will still be known as basic life support

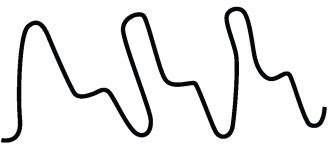
POSITIONING OF RESCUE TEAM

- Team leader stands on foot-end of patient
- 1 Doctor/Nurse stands on head-end of patients to ensure airway – Bag & mask or Bag & tube
- 2-3 rescuers will stand on left side of patient
 - Each gives chest compression for 2 mins & move away
- 1 Doctor/Nurse stands on right side of patient to secure an I/V line



HOW IS ACLS DIFFERENT FROM BLS?

- ACLS involves use of I/V drugs, whereas BLS does not
- In ACLS, doctor decides whether DC shock to be given or not, whereas in BLS, AED decides whether to deliver a shock

SHOCKABLE RHYTHM	NON-SHOCKABLE RHYTHM
VF 	Asystole 
VT 	PEA 



- Q. Q. A 50-year-old female collapsed to the ground while walking. You happened to be a bystander and approached the patient. On examination, a pulse was felt, but breathing efforts were absent. What is the next step in managing this patient?
- a. One cycle of cardiopulmonary resuscitation in 2 minutes
 - b. 10-12 breaths/minute or 1 breath every 6 seconds
 - c. Monitor the patient until the emergency services arrive
 - d. Automated External Defibrillator

Ans. (b)

- Q. During a cardiopulmonary resuscitation (CPR) scenario, what is the recommended rate for chest compressions, and how often should rescue breaths be administered in accordance with the standard compression-to-ventilation ratio?
- a. Chest compressions @80-100/min with 2 rescue breaths every 15 compressions
 - b. Chest compressions @120-140/min with 2 rescue breaths every 20 compressions
 - c. Chest compressions @100-120/min with 2 rescue breaths every 30 compressions
 - d. Chest compressions @60-80/min with 2 rescue breaths every 10 compressions

Ans. (c)

- Q. During the use of an Automated External Defibrillator (AED), what is the standard energy level, measured in joules, delivered in a direct current (DC) shock when the device detects a shockable rhythm?
- a. 50 Joules
 - b. 100 Joules
 - c. 150 Joules
 - d. 200 Joules

Ans. (d)



PULSELESS ELECTRICAL ACTIVITY

TREATABLE CAUSES OF CARDIAC ARREST

00:00:32

5 H's	5 T's
Hypoxia	Toxins
Hypovolemia	Tamponade
Hypokalemia/Hyperkalemia	Tension pneumothorax
Hydrogen Ion excess	Thrombosis: Pulmonary Embolism
Hypothermia	Thrombosis: Coronary event (MI)

HOW TO REMEMBER THIS?

WITH RESPECT TO HEART		WITH RESPECT TO LUNGS
Hypovolemia (Gunshot injury or aortic aneurysm rupture)	NOT ENOUGH	Hypoxia • T/t: Noninvasive ventilation, high flow nasal cannula, intubation
Cardiac Tamponade (Lung cancer - M/c: Oat cell cancer) • T/t: Pericardiocentesis o For trauma causing hemoperitoneum- Emergency resuscitative thoracotomy	SQUEEZED	Tension pneumothorax (one of the important causes - Barotrauma) • T/t: Wide bore needle in 5 th intercostal space • Absent breath sounds (differentiate it from cardiac tamponade)
Myocardial Infarction	KILLED	Pulmonary Embolism • Coronary artery thrombosis • Deep vein thrombosis → Retrograde spread of clot in pulmonary artery

REMAINING CAUSES OF PEA

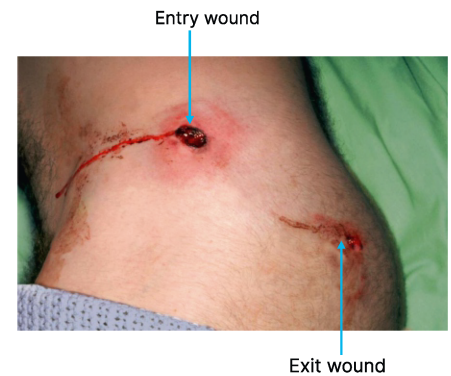
MNEMONIC - HEAP

- Hypothermia
- Electrolyte imbalance
 - o $\uparrow K^+$ -Cardiac arrest (T/t - Calcium gluconate)
 - o $\downarrow K^+$ -Torsades de pointes (T/t: KCl correction)
- Acidosis
- Poisoning
 - o E.g., TCA toxicity → Triggers development of arrhythmia

SHOCKABLE RHYTHM	NON-SHOCKABLE RHYTHM
- Unstable Ventricular Tachycardia (VT) - Survival: 75% - Ventricular Fibrillation - Survival: 25-40%	- Pulseless Electrical Activity (PEA) - Survival: 11% - Asystole - Survival: 0-2%

EMPTY HEART

- Gunshot injury: Extensive blood loss
 - Empty hear beating
 - Death occurs within 15 to 30 minutes without a massive blood transfusion/IV fluids
- Abdominal aorta aneurysm rupture d/t Atherosclerosis (in smokers)
 - AAA >5cm can spontaneously rupture → Hypovolemic shock

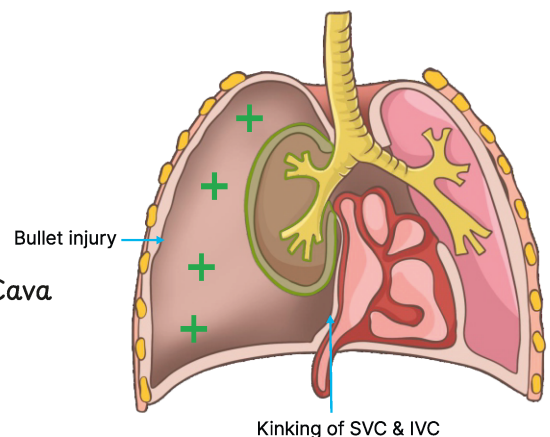


PRESSURE ON THE LUNG (TENSION PNEUMOTHORAX)

- Atmospheric Air = 760 mmHg
- Normal Intrapleural Pressure is negative
- Bullet Injury: Large Pressure Difference
 - ↓
 - Collapse of same side Lung
 - ↓
 - Heart & Tracheal Deviation to Opposite Side
- Air at positive pressure → Kinking of Superior & Inferior Vena Cava

↓ Venous Return to the Heart
 ↓ Cardiac Output
 ↓ Blood Pressure

- Tension pneumothorax: Example of obstructive shock



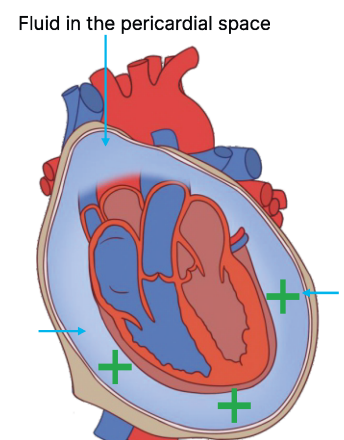
PRESSURE ON HEART

- Obstructive shock - E.g., Cardiac Tamponade (Non pulsatile elevated JVP, Beck's triad)
- Echocardiography: Diastolic collapse of ventricles
 - Oat cell cancer or Tuberculosis

↓
 Intracardiac space pressure > left ventricular end-diastolic pressure & right ventricular end-diastolic pressure

↓
 Impaired filling of chambers

- Management for medical causes like TB, lung cancer - Pericardiocentesis
 - Trauma: Emergency resuscitative thoracotomy





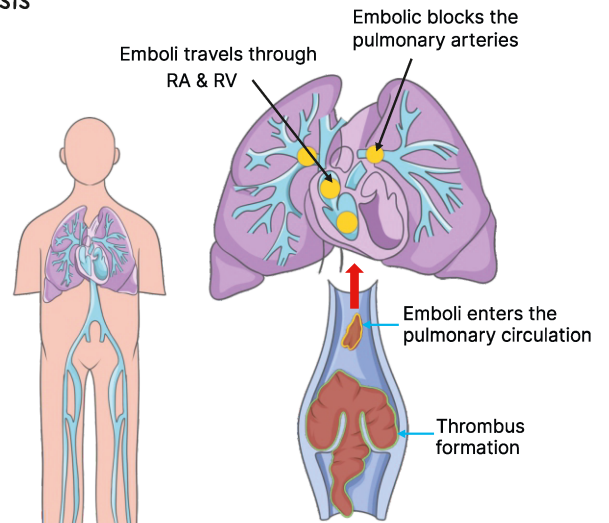
- Coronary artery thrombosis: Atherosclerosis in coronary artery
- Fissure fissure (platelet stick on it) → Clotting & Thrombosis

Myocardial Infarction

↓
Stunned Myocardium

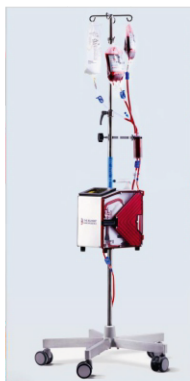
↓
Cardiogenic shock (leading cause of death)

- Pulmonary artery thrombosis: Deep vein thrombosis (DVT)
 - Hypercoagulable state:
 - Pregnancy
 - Factor V Leiden mutation
 - Joint implant surgery
 - Cancer surgery



INTERVENTIONS

00:14:45



Rapid Infusion with warm saline for hypovolemia & hypothermia



Pericardiocentesis



Wide-bore needle is inserted into the 5th intercostal space, anterior to mid-axillary line



Air blanket for hypothermia

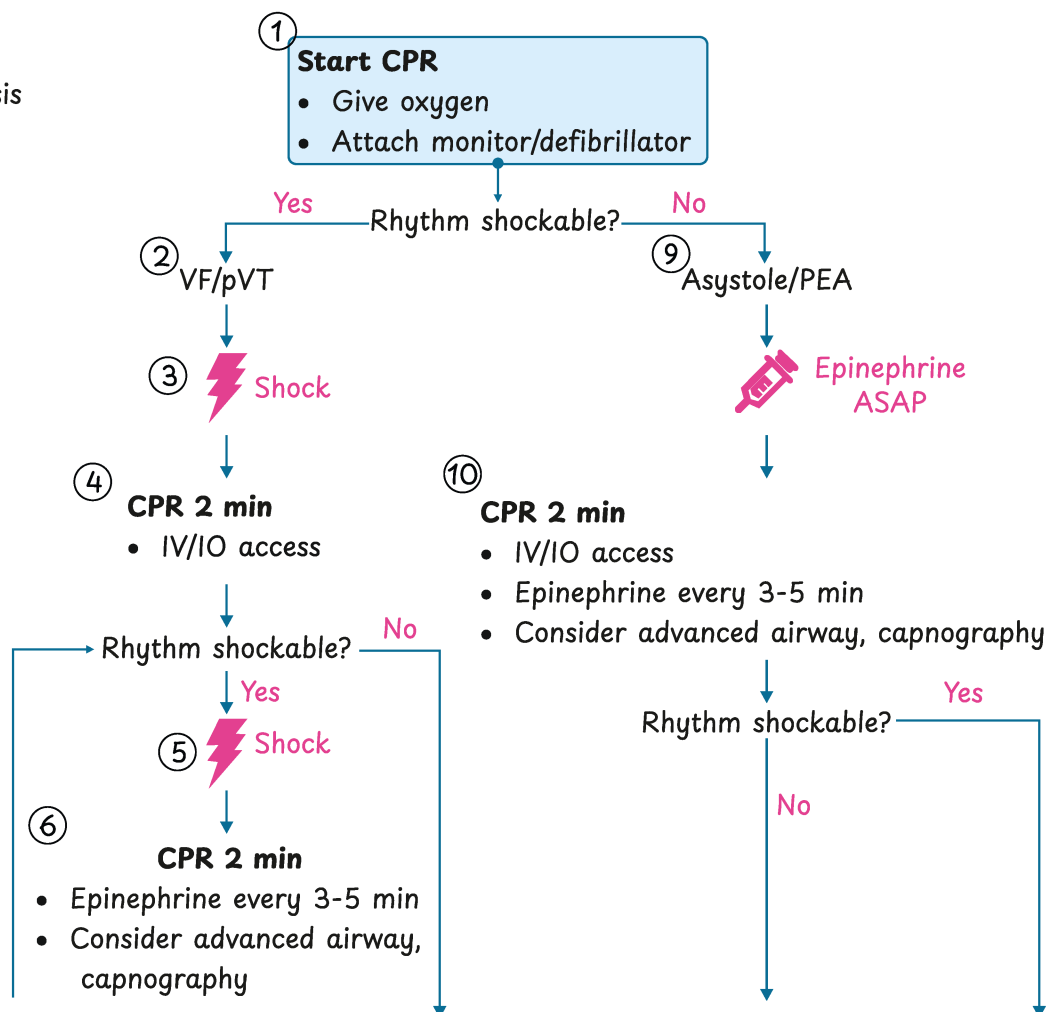
Hypoxia. Oxygen-based therapies • Non-invasive ventilation • Non-rebreather mask • High-flow nasal cannula	Toxins • Antidotes
Hypovolemia • Massive blood transfusion by wide-bore cannula • I.V. Fluids	Tamponade • Pericardiocentesis • Emergency resuscitative thoracotomy
Hypokalemia • Potassium chloride (KCl) infusion Hyperkalemia • Calcium gluconate • Insulin drip (lower value of potassium in blood)	Tension pneumothorax • Wide-bore needle insertion • Tube thoracostomy with underwater seal drainage
Hydrogen Ion excess • Soda bicarbonate	Thrombosis: Massive Pulmonary Embolism • Alteplase
Hypothermia • External rewarming- Air blanket	Thrombosis: Coronary event (MI) • Percutaneous coronary interventions (PCI) • Alteplase



Q. What is the first intervention for PEA?

- a. DC shock
- b. Epinephrine
- c. Check pulse
- d. Pericardiocentesis

Ans. (b)





30.9 ADVANCED CARDIAC LIFE SUPPORT

- DC shock – terminates **tachyarrhythmia**
- DC shock **C/I** in asystole/ flat line ECG

ACLS

00:02:21

- Use of emergency drugs
 - I/V or intraosseous access to administer drugs
 - Adrenaline
 - Amiodarone/ Lignocaine
 - Atropine
- Advanced airway (intubation)
- Decision to deliver or not deliver shock made by trained personnel
 - Non shockable rhythm - Asystole/ Pulseless electrical activity
 - Shockable rhythm - Ventricular fibrillation/ Pulseless ventricular tachycardia
- Type of DC shock decided by trained personnel
 - Cardioversion, Defibrillation
- Energy delivered by DC shock decided by trained personnel
 - **120 J – 200 J (Biphasic DC shock)**

CARDIOVERSION VS DEFIBRILLATION

00:08:45

CARDIOVERSION	DEFIBRILLATION
 • Delivers impulse at peak of 'R' wave • Synchronized DC shock	 • Non synchronized DC shock • Used in Ventricular fibrillation, Pulseless ventricular tachycardia

SHOCKABLE vs NON-SHOCKABLE RHYTHMS

SHOCKABLE RHYTHM	NON-SHOCKABLE RHYTHM
Ventricular fibrillation 	Asystole • Repeated Adrenaline (Asystole converted to ventricular fibrillation, DC shock can then be given) • DC shock – C/I



Pulseless Ventricular tachycardia



Pulseless electrical activity

- Low voltage ECG - cardiac tamponade



AGONAL RHYTHM

00:18:26

- Rhythm just before patient goes into cardiac arrest



REVERSIBLE CAUSES (5-H & 5-T)

5 H	5 T
- Hypovolemia - T/t - Massive transfusion - Hypoxia - T/t - O₂ via endotracheal intubation - H⁺ excess - Metabolic acidosis - T/t - Soda bicarbonate - Hyperkalemia/ Hypokalemia - K ↑ > 8.0 meq - Cardiac arrest - K ↓ < 1.5 meq - Torsades de pointes - Hypothermia - T/t - Rewarming	- Toxins - Tamponade - Tension pneumothorax - T/t - Wide bore needle at 5th ICS (mid axillary line) - Thrombosis in heart - M.I - Thrombosis in lung - Pulmonary embolism

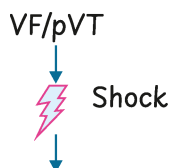
ALGORITHM IN ACLS

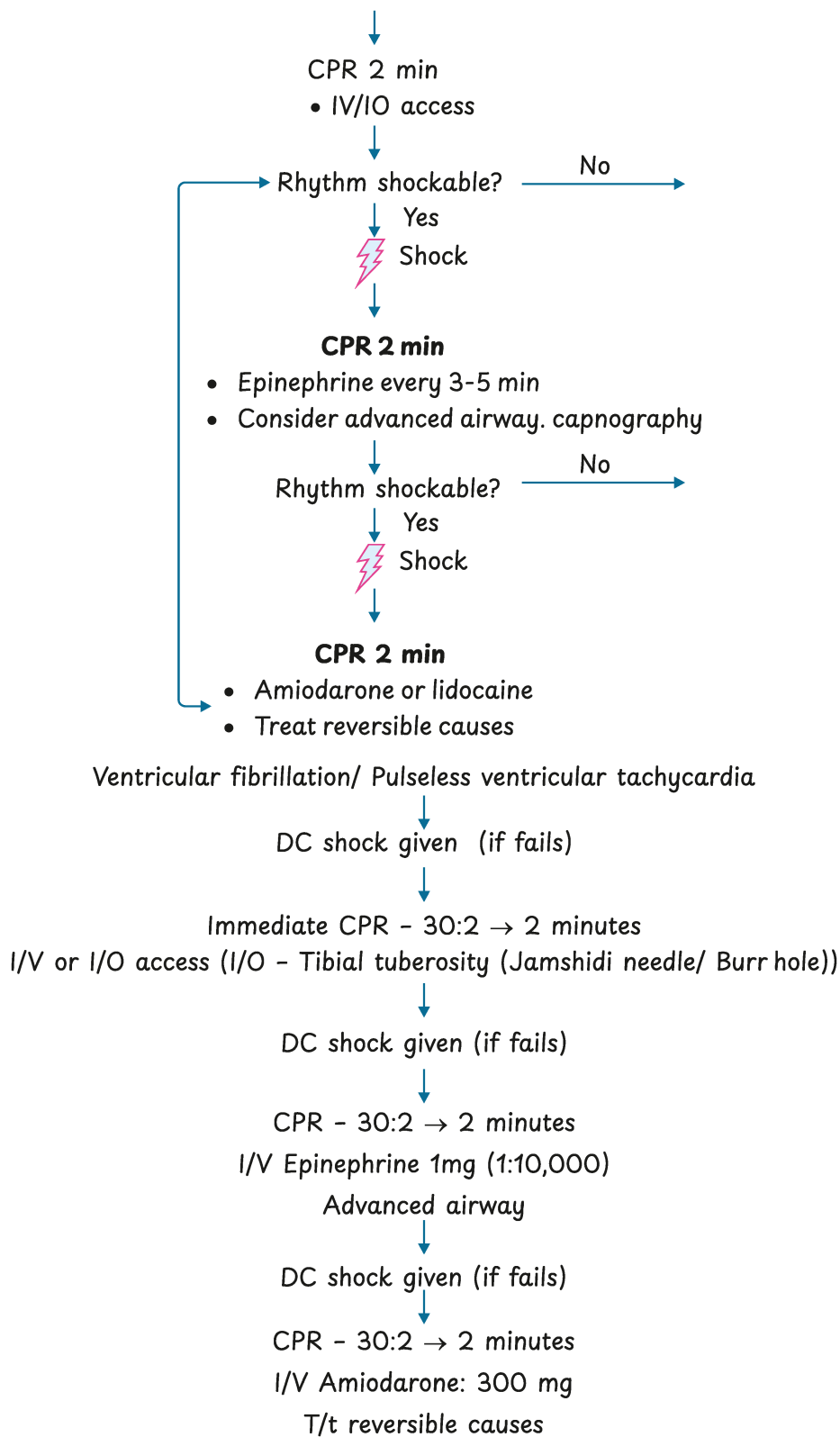
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- Initiate CPR
- Give O₂
- Attach defibrillator

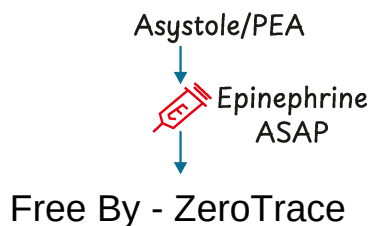
VENTRICULAR FIBRILLATION, PULSELESS VENTRICULAR TACHYCARDIA: DEFIBRILLATION

- 3 attempts of defibrillation can be given





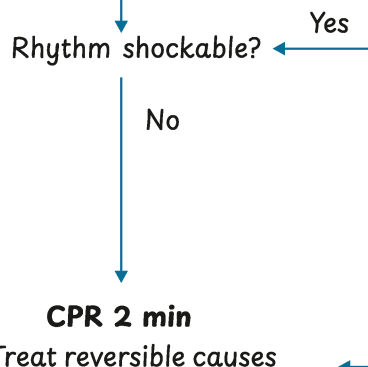
ASYSTOLE/ FLAT LINE ECG/ PULSELESS ELECTRICAL ACTIVITY



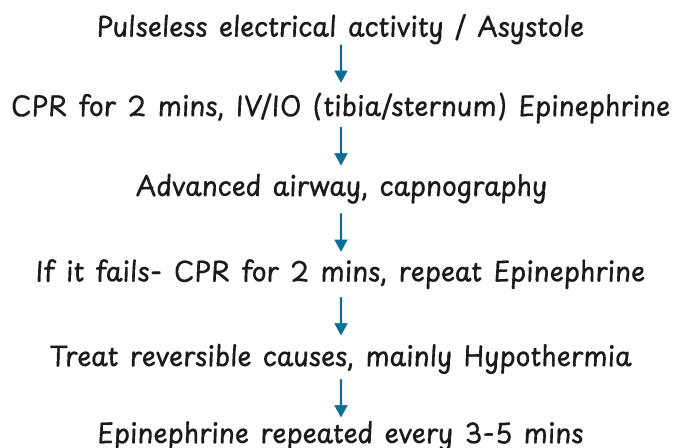


CPR 2 min

- IV/IO access
- Epinephrine every 3-5 mins
- Consider advanced airway, capnography



- Epinephrine : **ASAP** (repeat 3-5 mins)
- CPR : **2 min**
- I/V or I/O access: Epinephrine
- Consider advanced airway (endotracheal intubation)
 - **10/min** respiratory rate

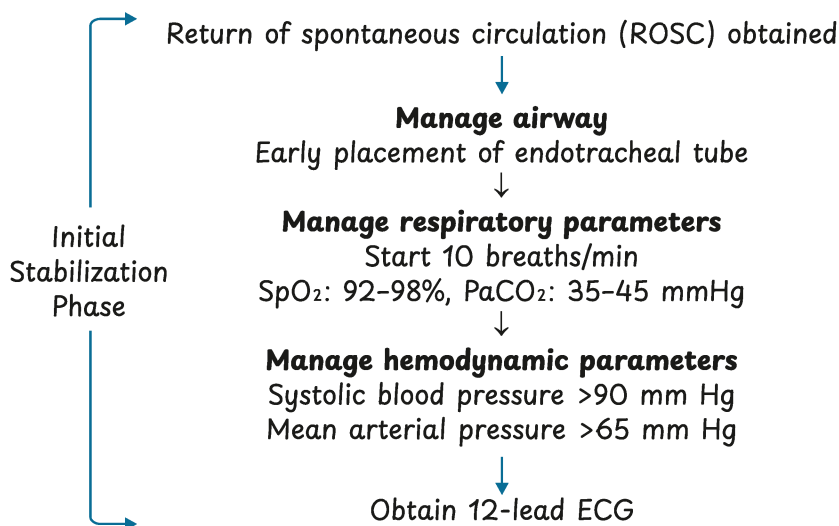


SHOCK ENERGY FOR DEFIBRILLATION	DRUG THERAPY	ADVANCED AIRWAY
• Biphasic: Manufacturer recommendation (eg, initial dose of 120-200J); if unknown, use maximum available. • Second and subsequent doses should be equivalent, and higher doses may be considered. • Monophasic: 360 J	• Epinephrine IV/IO dose: 1 mg every 3-5 minutes • Amiodarone IV/IO dose: First dose: 300 mg bolus. Second dose: 150 mg. (or) Lidocaine IV/IO dose: First dose: 1-1.5 mg/kg. Second dose: 0.5-0.75 mg/kg	• Endotracheal intubation or supraglottic advanced airway • Waveform capnography or capnometry to confirm and monitor ET tube placement • Once advanced airway in place, give 1 breath every 6 seconds (10 breaths/min) with continuous chest compressions

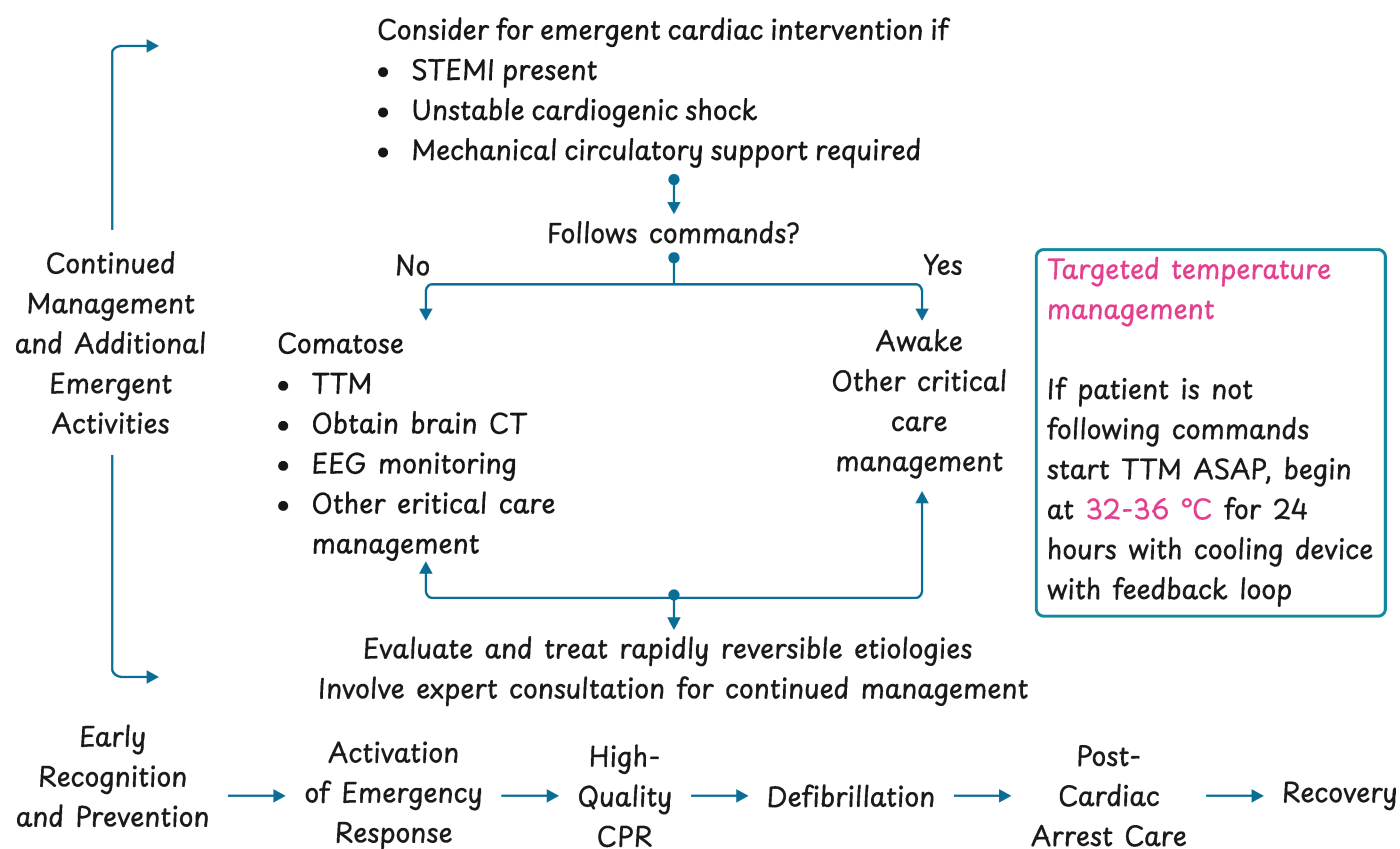


- Ventricular fibrillation/ pulseless ventricular tachycardia and Asystole → Epinephrine
- Pulseless ventricular tachycardia → Amiodarone

POST CARDIAC ARREST ALGORITHM



- STEMI present → PCI
- Unstable cardiogenic shock not responding to vasopressor → Mechanical circulatory support required → Impella/ Intra aortic balloon pump



- Therapeutic hypothermia - cerebro and cardio **protective**



Q. You are working in CCU when your patient of STEMI who was started on thrombolysis develops pVT. Which of the following is best step in management of this patient?

- a. Check pulse, remove ECG leads and start chest compressions
- b. Check scene safety
- c. Defibrillation
- d. Adenosine for chemical cardioversion

Ans: (c)



MECHANICAL VENTILATION

BASELINE PARAMETERS

00:00:36

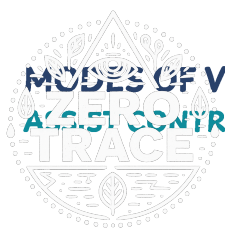
- Modes – ACMV/SIMV

PARAMETERS	NORMAL VALUE	VARIATION
Tidal Volume	6-8 ml/kg of lean body weight	ARDS: Ventilate at lower volumes – 5 to 6ml/kg of lean body weight to prevent barotrauma
Respiratory rate	12-16/min	Raised ICP- Elective hyperventilation
Peak Inspiratory Pressure	• <35cm H₂O • Plateau pressure – <30cm H₂O	
Inspiratory-Expiratory ratio (I/E ratio)	1:2	• Severe acute asthma & COPD – 1:3 • ARDS – 2:1 (Inverse ratio ventilation)
Fraction of Inspired Oxygen (FiO ₂)	1.0 (100% oxygen) ↓ 0.6 (60% oxygen) Initial target – • pO₂ > 75 mmHg • SpO₂ >90%	
Peak End Expiratory Pressure (PEEP)	5cm H ₂ O	• If PEEP >5cm H₂O used – ↓ venous return & Hypotension • Low PEEP used in – ○ Tension pneumothorax ○ Massive pulmonary embolism • High PEEP used in ○ ARDS

Important Information

Mechanical Ventilation in ARDS

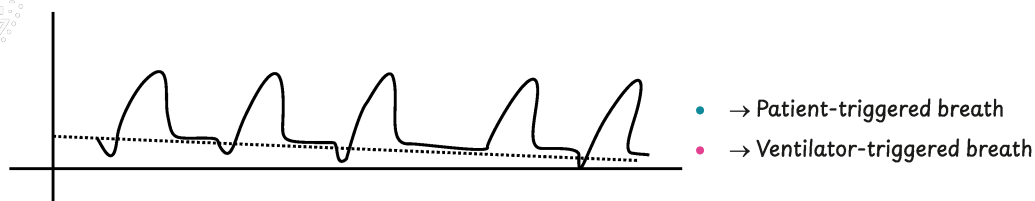
- Low volume ventilation (Mortality reducing step)
- Inverse ratio ventilation
- PEEP >5cm H₂O
- Prone position



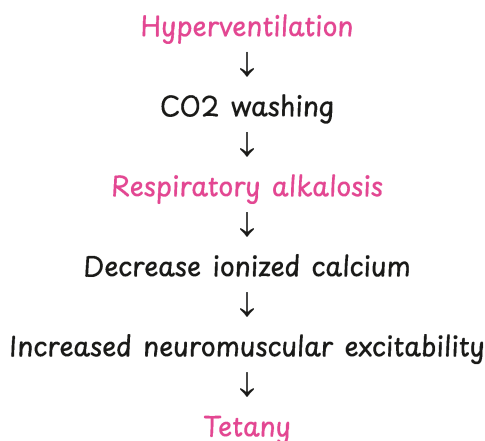
MODES OF VENTILATION

ASSIST-CONTROLLED MECHANICAL VENTILATION (ACMV)

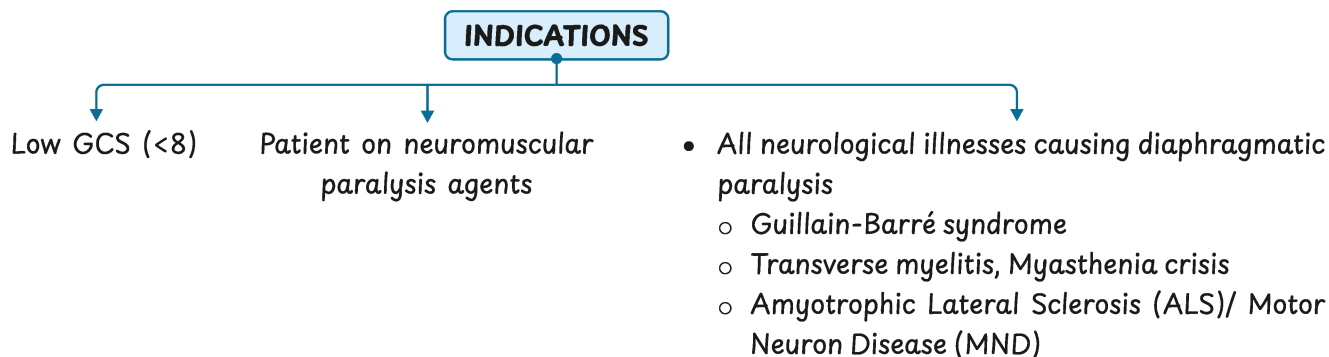
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- Patient exhibits breathing effort & ventilator provides assists in breathing
- Patient should exhibit breathing effort at least once every 5 seconds
- In absence of breathing effort, ventilator provides backup breath
- M/c mode of ventilation used
- S/E -

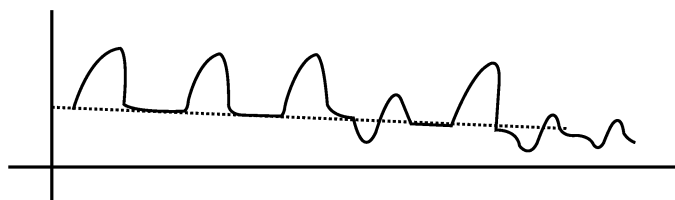


INDICATIONS



SYNCHRONIZED INTERMITTENT MANDATORY VENTILATION (SIMV)

- Every 5 seconds, breath is provided
- No assist during patient-triggered breaths
- If ventilator-triggered breath and patient-triggered breath occur simultaneously, ventilator will be inhibited, allowing patient to breathe independently
- Use - Weaning off ventilator
- S/E - If breathing effort of patient is insufficient → Hypoventilation

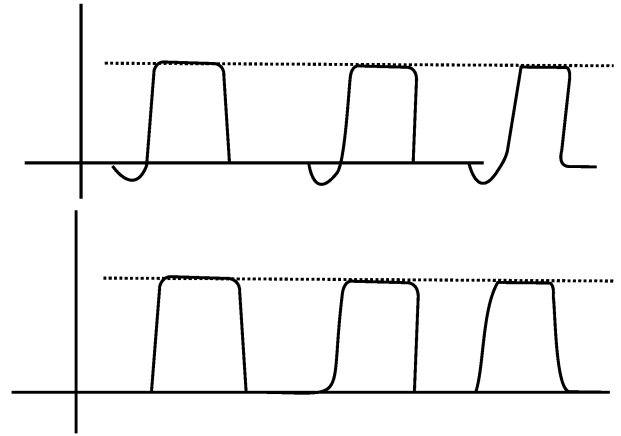


PRESSURE SUPPORT VENTILATION

- Patient triggered breaths to pre-set pressure
- Use - Weaning off ventilator

PRESSURE CONTROLLED VENTILATION

- Time triggered to pre-set pressure
- Use - Barotrauma

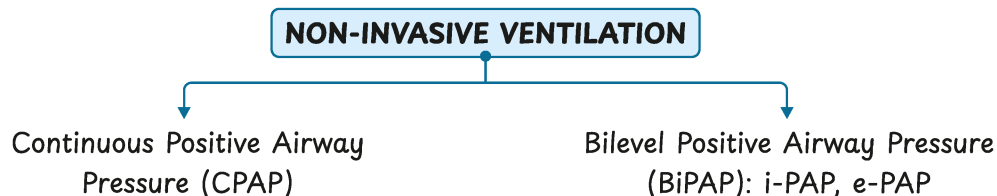


SCENARIOS WITH ABG PARAMETERS

	SCENARIO-1	SCENARIO-2
pH	7.4	7.2
pO ₂ (mm Hg)	60	100
pCO ₂ (mm Hg)	40	60
DIAGNOSIS	Acute Hypoxemic failure	Acute Hypercapnic failure
VENTILATOR PARAMETERS TO BE CHANGED	• ↑ FIO₂ • ↑ PEEP (↑ recruitment of alveoli)	• ↑ Respiratory rate (↑ minute ventilation as $MV = V_T \times RR$)

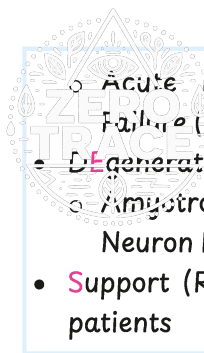
NON-INVASIVE VENTILATION

00:23:50



- MOA -
 - Pressure support without intubation
 - Helps to unload lungs

INDICATIONS	C/I
• Hypercapnic respiratory failure ○ Acute exacerbation of COPD ○ ABG - pH - 7.25 or less ○ Respiratory acidosis or CO₂ narcosis ○ Asterixis • Obstructive Sleep Apnea, • Obesity Hypoventilation Syndrome, Central Sleep Apnea • Pulmonary edema (Cardiogenic)	• Face trauma/ facial burns • Airway obstruction (upper) • Coma • Excessive secretions • Pneumothorax (tension) • Aspiration risk • Inability to cooperate/ hemodynamic instability ○ Cardiogenic shock/ MI ○ Upper GI bleed • Non-fitting face mask



- Acute Decompensated Congestive Heart Failure (ADCHF)/ Acute Heart Failure (AHF)
- **D**egeneration of neurons
- Amyotrophic Lateral Sclerosis (ALS)/ Motor Neuron Disease (MND)
- **S**upport (Respiratory) in immunocompromised patients

ADVANTAGES

- ↓ chances of laryngotracheal trauma/ ventilator-associated pneumonia

STEP-UP OXYGEN THERAPY IN SARS-COV-2

Step 1:

Non-rebreather mask (NRBM)



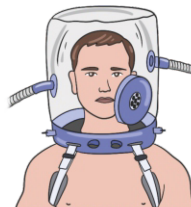
Flow of oxygen: 15L/min

↓ Fails

Non-invasive ventilation

Face Mask

Helmet Interface



↓ Fails

High-flow nasal cannula (HFNC)



Oxygen flow rate: 60L/min

↓ Fails

Tracheostomy + A.C.M.V.

Free By - ZeroTrace